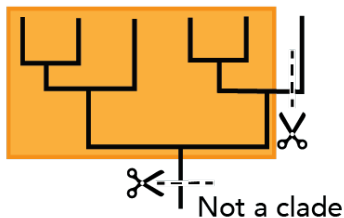
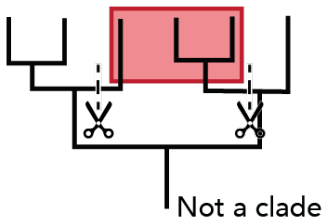
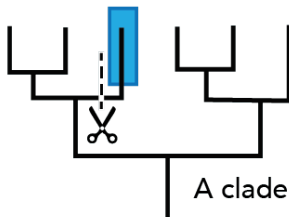
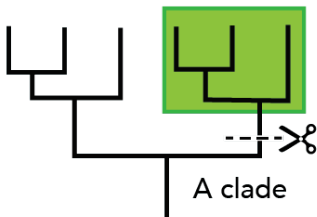




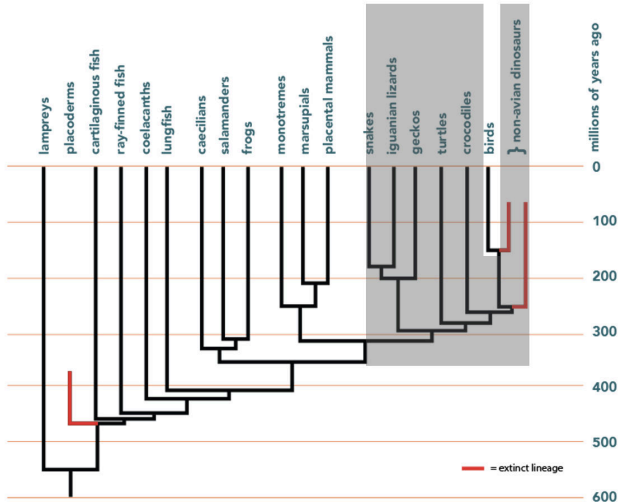
Avian Phylogenetics

Dr. Ethan Linck
Office: 305 Lewis Hall

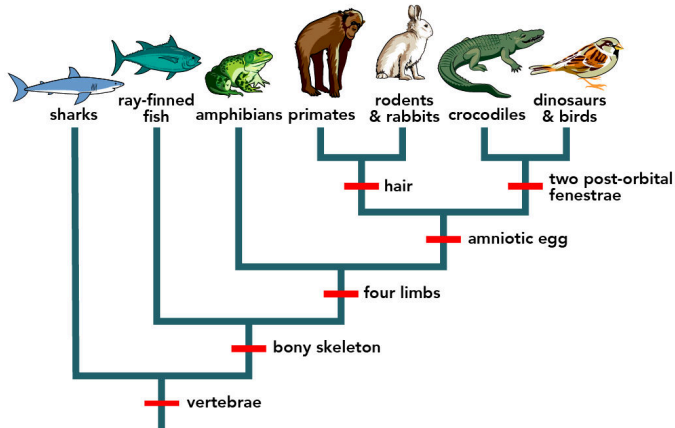
Taxonomic groups are (usually) monophyletic



Taxonomic groups are (usually) monophyletic



“Lower” (narrowly defined) taxa share derived traits



- ▶ Basal traits: evolved earlier; shared by all
- ▶ Derived traits: evolved later; define subclades

Morphological phylogenetic systematics

- Do named taxa share derived traits that reflect *ancestry* (not convergent natural selection)?

MONOPHYLY OF THE PASSERIFORMES: TEST OF A PHYLOGENETIC HYPOTHESIS

ROBERT J. RAIKOW

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Pittsburgh, Pennsylvania 15260 USA*

ABSTRACT.—The order Passeriformes was largely defined by 19th century anatomists on the basis of phenetic similarities. The purpose of this study was to determine whether the group is monophyletic in the strict contemporary sense. Similarities used by previous workers were reanalyzed to determine whether or not they could be shown to be derived character states corroborating the hypothesis of passeriform monophyly. Of 18 traditional taxonomic characters analyzed, none refuted the hypothesis of passeriform monophyly, 13 failed to corroborate the hypothesis, and 5 did corroborate it. These are the aegithognathous palate, the “passerine” tensor propatagialis brevis, the bundled spermatozoa with coiled head and large acrosome, the enlarged hallux, and the type VII deep plantar tendons.

To this analysis is added new information from the hind limb musculature. Previous knowledge of oscine musculature is augmented by new information from representative suboscines. The division of M. pubo-ischio-femoralis into Pars cranialis and Pars caudalis is confirmed for suboscines, distinguishing the entire order Passeriformes from nonpasserine birds. In the foot, the loss of a set of intrinsic muscles of the forward digits, previously known from oscines, is also confirmed for suboscines. These limb-muscle characters supplement the analysis of traditional characters in corroborating the hypothesis that the order Passeriformes is monophyletic.

A functional analysis of the passerine foot shows that it is a derived mechanism specialized for perching, but with a reduction in the subtlety and variety of certain other movements. Received 9 March 1981, accepted 3 August 1981.

Morphological phylogenetic systematics

► Do named taxa share derived traits?

JULY 1982]

Passeriform Monophyly

435

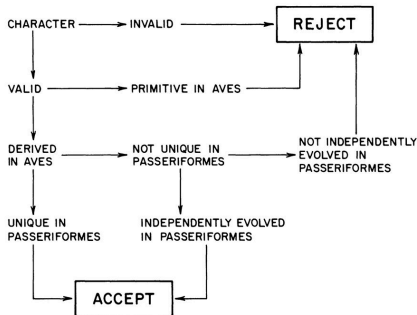


Fig. 1. A scheme for determining whether traditional phenetic characters are derived within Aves at the level of Passeriformes and whether they will therefore corroborate the hypothesis of passeriform monophyly. Characters that are *accepted* do corroborate the hypothesis; those that are *rejected* do not. See text for discussion.

(Raikow 1982)

Morphological phylogenetic systematics

- Do named taxa share derived traits?

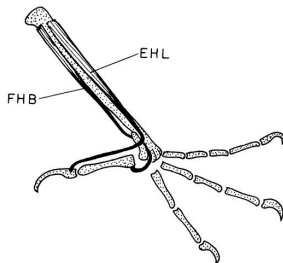


Fig. 3. Diagrammatic representation of the two intrinsic foot muscles generally found in passerine birds, *M. extensor hallucis longus* (EHL) and *M. flexor hallucis brevis* (FHB).

(Raikow 1982)

Morphological phylogenetic systematics

- Morphological methods established many groups!



Division 2

Order Palaeognathiformes

Suborder Tinami

Family Tinamidae

Suborder Ratiti

Infraorder Apteryges

Family Dinornithidae

Apterygidae

Infraorder Struthioness

Superfamily Casuariodea

Family Casuariidae

Dromiceidae

Superfamily Struthionioidea

Family Rheidae

Struthionidae

(Cracraft 1982)

When does morphology fail?

- ▶ Plumage can converge because of biotic interactions like mimicry
- ▶ E.g., Hairy Woodpecker (*Leuconotopicus villosus*) and Downy Woodpecker (*Dryobates pubescens*)

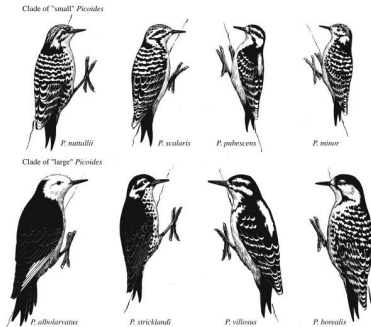
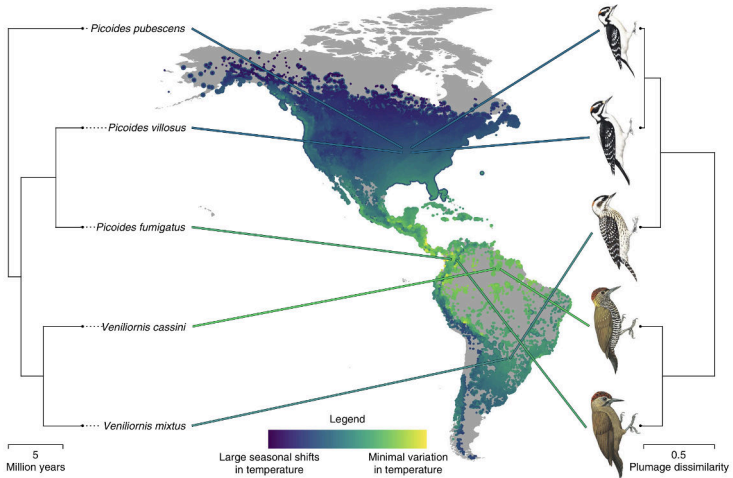


FIGURE 1. Sketches of North American "small" *Picoides*: *P. nuttallii*, *P. scalaris*, *P. pubescens*, *P. minor* (a Eurasian species); and "large" *Picoides*: *P. albolarvatus*, *P. stricklandi*, *P. villosus*, *P. borealis*. Note the extensive similarity in plumage between the distantly related species, *P. pubescens* and *P. villosus*. Drawings by John Megahan from study skins housed in the University of Michigan Museum of Natural History.

When does morphology fail?

- ▶ Genetic data confirms analysis of traits other than plumage in Hairy / Downy pair



Molecular phylogenetics

- DNA sequences provide millions of independent “traits”

t1	G	C	T	T	C	T	G	A	T	T	A	A	C	C	T	G	C	T
t2	G	C	T	T	C	T	G	A	T	T	T	C	T	C	T	G	C	C
t3	G	C	T	T	C	T	G	A	T	T	A	C	T	C	T	G	C	C
t4	G	C	T	T	C	T	G	A	C	T	A	G	T	C	T	G	C	T

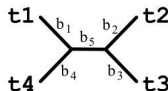
Site
pattern

GGGG
CCCC
TTTT
AAAA
TTTC
ATAA
ACCG
CTTT
TCCT

Count

3
4
5
1
1
1*
1*
1*
1*

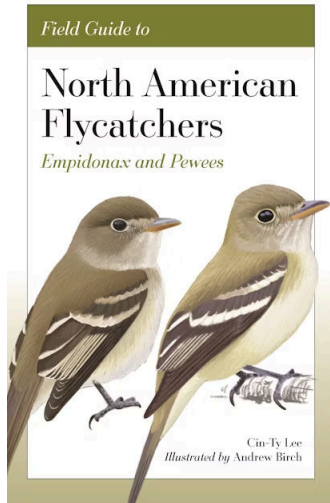
$$L(H|R) = \gamma P(R|H)$$



The tree (with branch lengths b_i) is the hypothesis (H) and the site patterns are the data (R)

$$d_{12} = -\frac{3}{4} \ln\left(1 - \frac{4}{3}p_{12}\right)$$

Phylogenetics helps evaluate cryptic diversity



(Lee & Birch 2023)

Case Study: Pacific Slope / Cordilleran Flycatcher

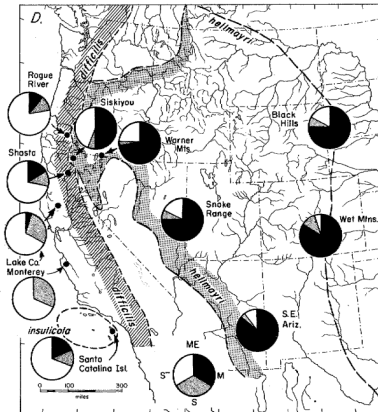
- ▶ Highly similar Western US flycatchers



(*Empidonax difficilis* / *Empidonax occidentalis*)

Johnson split “Western” flycatcher

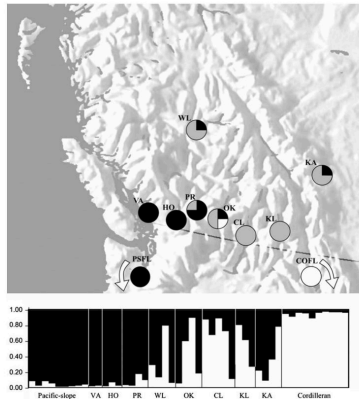
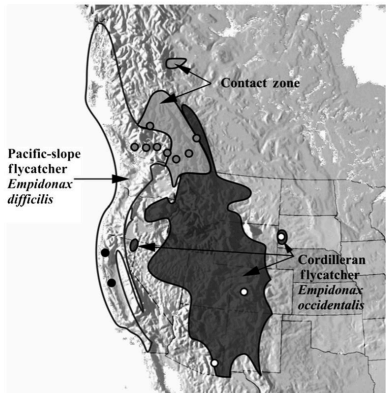
- Early genetic data + variation in vocalization



(Johnson & Marten 1988)

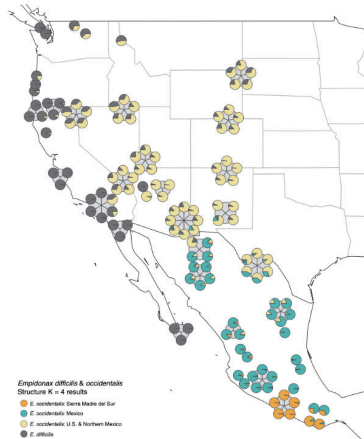
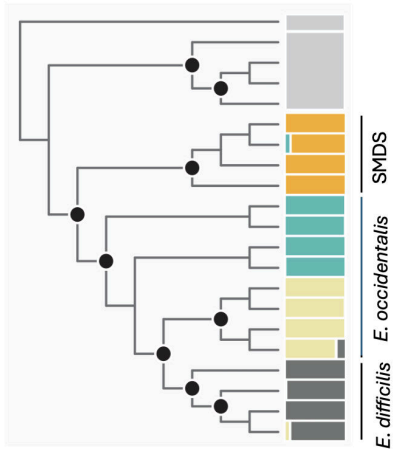
But Pacific Slope and Cordilleran Flycatcher hybridize!

- Contact zone in NW Montana / BC



(Rush et al. 2009)

Is Cordilleran Flycatcher *E. occidentalis* monophyletic?



(Linck et al. 2019)

Proposal to American Ornithological Society to “lump”

Treat Pacific-slope Flycatcher *Empidonax difficilis* as conspecific with Cordilleran Flycatcher *E. occidentalis*

Note: some passages in this proposal are directly borrowed or modified from Hopping (2022).

Background:

Previously considered conspecific, Pacific-slope Flycatcher *Empidonax difficilis* (*sensu stricto*) and Cordilleran Flycatcher *E. occidentalis* were split from Western Flycatcher *E. difficilis* (*sensu lato*) by the American Ornithologists' Union (AOU 1989), based primarily on Ned K. Johnson's impressive 1980 monograph concerning the *Empidonax difficilis-flavescens* complex (Johnson 1980), as well as the conclusions drawn by Johnson & Marten (1988). The stated justification for the split was that the two forms “differ in vocalizations and allozyme frequencies and are sympatric in the Siskiyou region of northern California” (AOU 1989).

Despite the extensive dataset featured in his original monograph, Johnson (1980:110–115) concluded at the time that the only clear species break in the complex was between Western Flycatcher *E. difficilis* (*sensu lato*) and *E. flavescens* (Yellowish Flycatcher), and he recommended that interior (*E. d. hellmayri*) and coastal (*E. d. difficilis*) forms were best left tentatively as “megasubspecies” (Amadon & Short 1976). Following analysis of allozyme frequencies, reinterpretation of the earlier samples, and the collection of 17 new specimens (including three pairs from the Siskiyou region of northern California, where the two forms were suspected to occur in sympatry) in the summer of 1981, Johnson revised this stance (Johnson & Marten 1988), leading to his successful push for the split (AOU 1989).

(Chesser et al. 2023)

Western Flycatcher lives again. . . for now



Submit Explore My eBird Science About News

DATA FOR:  World

Change Species  Pine Flycatcher

Passeriformes > Tyrannidae

Western Flycatcher

Empidonax difficilis

SEEN PHOTO AUDIO YEAR



© Liam Singh eBird 567977060 Macaulay Library ML 236799211

Identification

POWERED BY MERLIN

Small, yellow-olive flycatcher. Brighter yellow than most other Empidonax flycatchers. Look for bold teardrop-shaped eyering that comes to a point behind the eye, and fluffy, peaked crown. Breeds in coniferous and mixed woodlands, often near water; migrants and wintering birds can be found in a variety of wooded or scrubby habitats. Extensive range across western North America, breeding from southeast Alaska to Oaxaca. Winters mainly in west and central Mexico. Listen for squeaky, high-pitched song and calls. In winter, most often detected by unobtrusive high-pitched "tink" call. Formerly split into two species, Cordilleran and Pacific-slope Flycatcher.

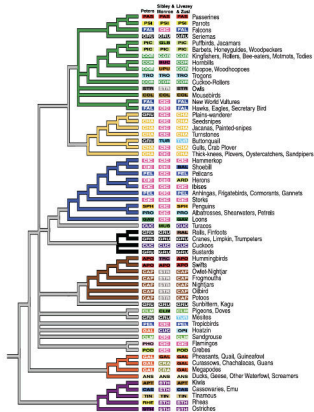
What could resurrect PSFL and COFL?



- a) more genomic data
- b) samples from a new part of COFL's range
- c) evidence genetic markers in Linck et al. were influenced by selection
- d) a new species concept
- e) all of the above

Avian phylogenomics: Hackett et al. 2008

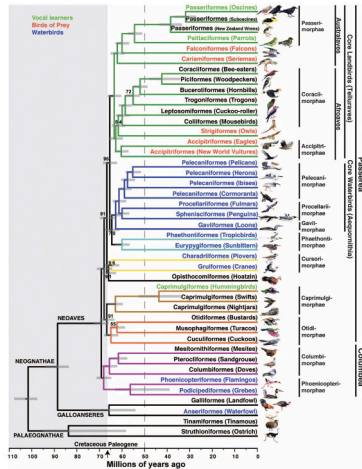
- ▶ Passerines sister to parrots
- ▶ Flamingos sister to grebes



(169 species, 32 kilobases, Hackett et al. 2008)

Avian phylogenomics: Jarvis et al. 2014

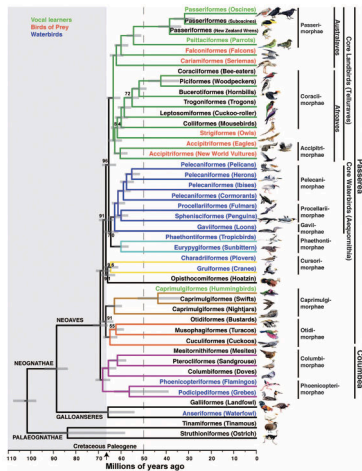
- ▶ Ancestor of core land birds a bird of prey
- ▶ Neoaves diversifies at K-Pg Boundary



(48 species, whole genomes, Jarvis et al. 2014)

Avian phylogenomics: Jarvis et al. 2014

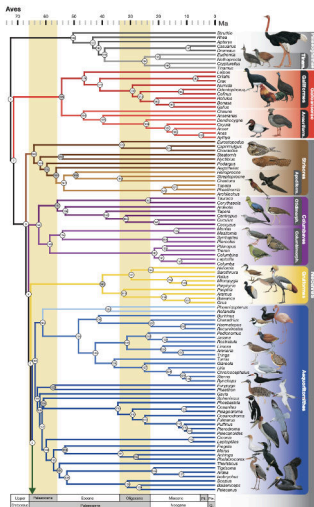
- ▶ Ancestor of core land birds a bird of prey
- ▶ Neoaves diversifies at K-Pg Boundary



(48 species, whole genomes, Jarvis et al. 2014)

Avian phylogenomics: Prum et al. 2015

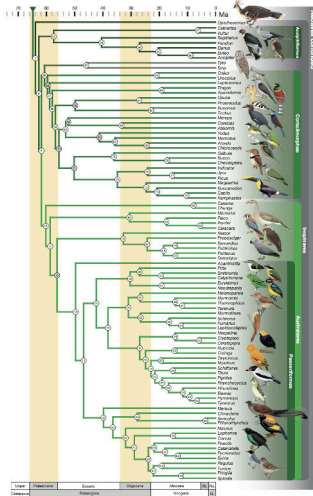
- ▶ No support for Columbea and Passerea
- ▶ Hoatzin sister to clade of landbirds within Neoaves



(198 species, 259 loci, Prum et al. 2015)

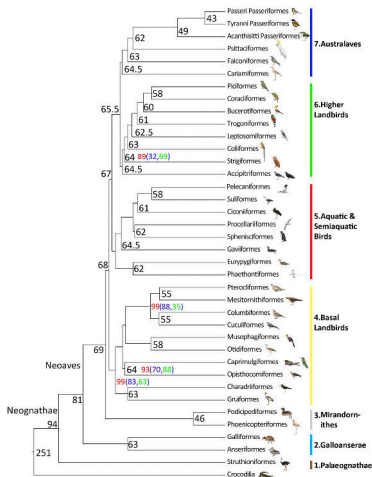
Avian phylogenomics: Prum et al. 2015

- ▶ No support for Columbea and Passerea
- ▶ Hoatzin sister to clade of landbirds within Neoaves



Avian phylogenomics: Kuhl et al. 2020

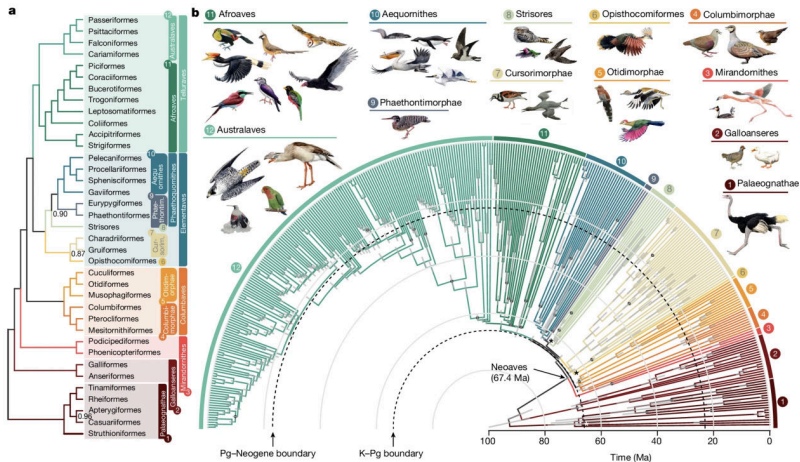
- ▶ Ancestor of core land birds a bird of prey
- ▶ Neoaves diversifies at K-Pg Boundary



(221 species, “untranslated regions”, Kuhl et al. 2020)

Avian phylogenomics: Stiller et al. 2024

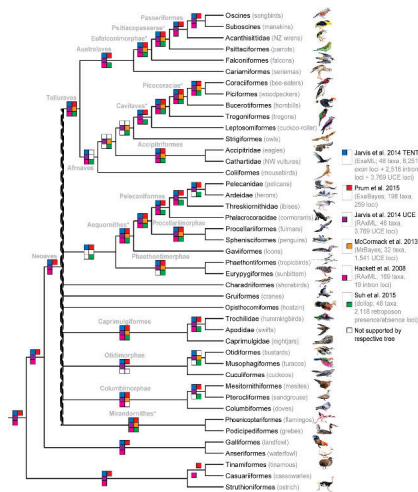
► 363 bird species & 63,430 intergenic loci



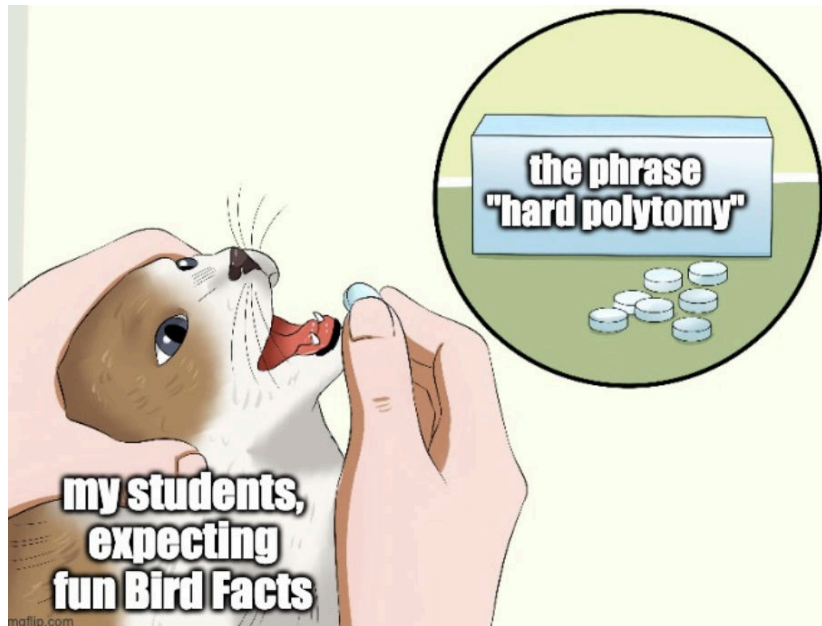
(Stiller et al. 2024)

A hard polytomy at the base of Neoaves?

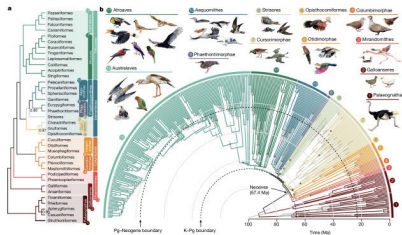
- Unresolvable because of 9 near-instantaneous speciation events



In conclusion...



What can we conclude about the avian tree of life?



- a) many lineages in Neoaves are only distantly related
- b) extinction of non-avian dinosaurs permitted rapid diversification of birds
- c) many familiar groups (e.g., Passerines) are monophyletic
- d) the Hoatzin is weird
- e) all of the above

Things to know

Name representative taxa from the following groups:

- ▶ Paleognathae, Neognathae, Galloanseres, and Neoaves
- ▶ Aequiornithes
- ▶ 'Orphaned' Neoaves (Camprimulgiformes, Columbiformes, Gruiformes, Hoatzin, others)
- ▶ Telluraves (Afroaves + Australaves)

Remember:

- ▶ How taxonomy and phylogenetics are related
- ▶ Relationships in the 'orphaned' Neoaves are hard to determine due to rapid radiation at K-Pg boundary
- ▶ Afroaves contains woodpeckers, kingfishers, owls, accipiters
- ▶ Australaves contains parrots, falcons, passerines

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